

XIE Yusen

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Education

- Sep 2023 - Present

Hong Kong University of Science & Technology (GZ)

ROAS (Doctorate)

Research Area: Computer Vision, Deep Learning CV, Multi-sensor Fusion SLAM, End-to-End Autonomous Driving
- Sep 2020 - Jun 2023

Beijing Information S & T University

IntelliSense Engineering (Master)

Research Area: Computer Vision, Deep Learning CV, Multi-sensor Fusion SLAM, Robot Localization and Navigation
- Sep 2016 - Jun 2020

Wuhan University of Technology

Geographic Information System (Bachelor)

Courses: Introduction to Geographic Information Systems, GIS Mapping and Information Extraction, Remote Sensing Principles and Image Processing, and accumulated rich programming experience

Publications

- Y. Xie, Z. Huang, K. Chen, L. Zhu, and J. Ma*, "MCGMapper: Light-Weight Incremental Structure from Motion and Visual Localization With Planar Markers and Camera Groups," *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024,
- Y. Xie, L. Deng, T. Sun* and B. Chen, "A4LidarTag: Depth-Based Fiducial Marker for Extrinsic Calibration of Solid-State Lidar and Camera," in *IEEE Robotics and Automation Letters (RAL)*, vol. 7, no. 3, pp. 6487-6494, July 2022.
- Y. Xie, T. Sun*, L. Deng and B. Chen, "Fast-robust book information extraction system for automated intelligence library," *2021 IEEE International Conference on Artificial Intelligence and Industrial Design (AIID)*, 2021.

Projects

- Aug 2021 - Present

Fully automatic aircraft jet bridge docking guidance system

Algorithm Leader

Role: Achiever and leader of camera-LiDAR fusion navigation algorithm

Objective: Utilize the stereo camera and solid-state LiDAR carried by the jet bridge, and through the principle of multi-sensor fusion, real-time estimate the door pose and complete the jet bridge navigation docking procedure

Completion: Successfully completed first fully automatic aircraft jet bridge guidance system in Asia. It has been developed in major airports nationwide with significant results.

Jun 2022 - Present

Fully autonomous navigation of a warehouse inventory robot

Algorithm Leader

Role: Map localization algorithm leader, navigation & control leader

Objective: Place multiple markers in the indoor warehouse, achieve global localization and real-time pose tracking of the car. Then controlling the autonomous movement of the robot to complete the unmanned inventory of the warehouse

Completion: Through the Road Network path management and pure pursuit path tracking algorithm, along with the robot control module, the inspection robot's functions can be independently completed, achieving the intelligent and flexible transformation of the warehouse. It has been developed in several warehouses nationwide

Nov 2020 - Present

Perception of library book information

Algorithm Leader

Establish an automated book information collection system composed of multiple cameras and photoelectric triggers, utilizing deep learning computer vision (e.g. detection, segmentation) to extract information from image data. The project has been running stably to date.

Honors & Awards

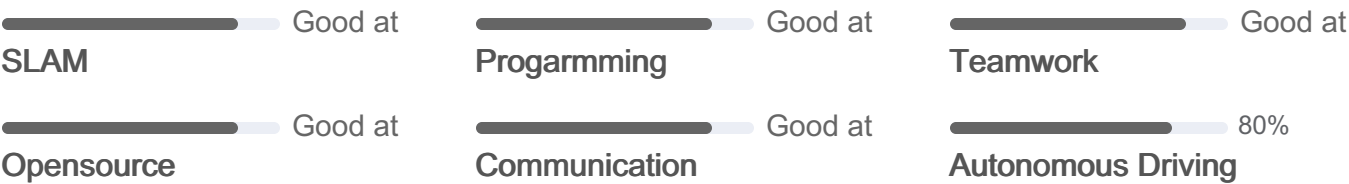
- National Scholarship for Graduate Students (0.2% Top)
- Outstanding Graduates in Beijing (1% Top)
- Excellent Master's Degree Thesis (1% Top)
- First-class Academic Scholarship for the 2021-2022 academic year (10% Top)
- Second-class Academic Scholarship for the 2020-2021 academic year (40% Top)

Skills

Professional Skills: CET-6, proficient in reading and writing academic papers, familiar with Linux environment,

Basic Knowledge: Mastery of basic principles and methods of image processing, computer vision, and SLAM, proficient in using deep learning for image processing

Teamwork: Abundant experience in algorithm development and programming practice, enthusiastic about using various technologies for algorithm design, practical experience in product implementation



Self-evaluation

- Algorithm Enthusiast
- Programming Enthusiast
- Brainstorming Enthusiast
- Hardworking
- Interesting
- Team Coordinator
- Energetic
- Self-Driven

谢玉森 YUSEN XIE

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教育背景

香港科技大学（广州） - 机器人与自主系统 - 博士	2023.09 - 2026.06
香港科技大学 - 电子信息和计算机工程 - 博士	2025.02 - 2025.07
北京信息科技大学 - 电子信息 - 硕士	2020.09 - 2023.06
武汉理工大学 - 地理信息系统 - 本科	2016.09 - 2020.06

个人简介

博士期间聚焦自动驾驶系统研究，主要研究 3DGS 优化、视觉感知算法（Depth Estimation）等。具备从算法原型设计（PyTorch）、嵌入式部署（C++/CUDA/ROS）到实车测试的全链路落地经验。熟悉 CV、SLAM、CUDA 加速及前沿感知方案研究。寻求端到端智能驾驶方案研发岗位，或计算机视觉相关岗位，推动前沿技术量产落地。

获奖荣誉

北京市优秀毕业生	2023.06
研究生国家奖学金	2022.10
北京信息科技大学优秀硕士学位论文	2023.06
硕士一等学业奖学金	2022.12
硕士二等学业奖学金	2021.12

项目经历

华为 2012 实验室中央媒体技术院 | 泊车场景的端到端 VLA 模型研究 2025.6-2026.6（预计）

项目技术：大模型/VLA

项目描述：构建面向泊车场景的端到端 VLA 模型，实现交互情形下可解释的决策推理与安全、高效的轨迹规划，旨在将真实泊车业务场景中的难例解决率提升至 80% 以上

项目职责和预期成果：

- 预计构建适用于 VLA 模型训练的泊车场景指令-微调数据集，涵盖 100K 量级真实行车车迹数据与视觉问答任务数据
- 预计构建多任务的 VLA 模型，基于多模态输入进行场景理解、决策推理、目标点生成和轨迹规划
- 预计设计多阶段的 VLA 模型训练方案并完成业务数据上的开环与闭环测试

考虑几何信息的多模态 3DGS 渲染

2025.5 - 至今

项目技术：3DGS/CUDA

项目描述：通过 RGB/Depth/Normal/Semantic 信息优化 3DGS 的生成

项目职责：

- 重新设计了考虑旋转 Rotation 和缩放 Scale 的 3DGS Depth 光栅化过程
- 实现了基于 CUDA 的从 Normal 损失到 3DGS 形状和旋转的反向传播链式求导过程
- 实现了 3DGS 对 Semantic 信息的前向渲染和反向传播
- 拓展了同时渲染 RGB/Depth/Normal/Semantic 信息的多模态查看软件

项目成果：实现了通过 RGB/Depth/Normal/Semantic 信息优化 3DGS 的生成，并在几何结构上完成了 3DGS 模型的高质量重建

融合 3DGS 的室外大场景图片级实时重建

2024.5 - 2024.9

项目技术：3DGS/Pose Estimation/CUDA

项目描述：融合 LiDAR-Camera-IMU SLAM 和 3DGS，实现室外场景高质量 3DGS 重建

项目职责：

- 设计了从单帧点云到精简 3DGS Representation 的 Gaussian Process 过程，实现冗余点云的 Gaussian 结构化
- 实现了结构化 3DGS 旋转 Rotation 和缩放 Scale 的快速参数化估计

3. 在多个室外 SLAM 数据集上均取得 SOTA 效果, 在 GPU 资源有限的情况下实现实时图片级大规模室外场景重建

项目成果: 高效耦合了 SLAM Pose Estimation 和 3DGS, 首次实现了室外大场景下的实时 3DGS 重建

博智林机器人研究院 | 多传感器融合三维重建框架

2023.09 - 2024.09

项目技术: ROS/PCL/Open3D/CPP

项目描述: 建筑机器人环境感知与高精度重建系统

项目职责:

1. 开发基于 iESKF 的多传感器融合算法, 实现 LiDAR-IMU 高质量三维重建
2. 纯激光雷达和 IMU 系统与三维扫描仪真值对比 RMSE<0.02 米
3. 设计自动化评估系统, 支持重建精度/完整性/时效性多维度评测

项目成果: 实现厘米级实时三维重建 (精度: 0.02 米)。支持 1W 点云/秒处理速度。

北京清瞳时代科技有限公司 | 仓储盘点机器人导航系统

2022.08 - 2023.06

项目技术: ROS/GTSAM/Cartographer

项目描述: 智能仓储无人化改造解决方案

项目职责:

1. 开发多标记物联合定位算法, 定位精度达 ± 0.02 米
2. 构建基于拓扑地图的路径规划系统, 支持动态避障
3. 优化路径控制算法, 路径跟踪误差 <0.05 米

项目成果: 提升盘点效率, 降低人工成本 99%。已在全国各大 1 万平方米示范仓库稳定运行。

发表刊物

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- [1] **Y. Xie**, Z. Huang, J. Jiao, D. Kanoulas, and J. Ma, "UniGS: Unified Geometry-Aware Gaussian Splatting for Multimodal Rendering," *Under Review*, 2025.
 - [2] **Y. Xie**, Z. Huang, J. Wu, and J. Ma, "GS-LIVM: Real-Time Photo-Realistic LiDAR-Inertial-Visual Mapping with Gaussian Splatting," *ICCV (CV TOP)*, 2025.
 - [3] **Y. Xie**, Z. Huang, S. Shen, and J. Ma, "Semi-SMD: Semi-Supervised Metric Depth Estimation via Surrounding Cameras for Autonomous Driving," *Under Review*, 2025.
 - [4] **Y. Xie**, Z. Huang, K. Chen, L. Zhu, and J. Ma, "MCGMapper: Light-Weight Incremental Structure from Motion and Visual Localization With Planar Markers and Camera Groups," *IROS (Robotics TOP)*, 2024.
 - [5] **Y. Xie**, L. Deng, T. Sun, and B. Chen, "A4LidarTag: Depth-Based Fiducial Marker for Extrinsic Calibration of Solid-State Lidar and Camera," *RAL (Robotics TOP)*, vol. 7, no. 3, pp. 6487-6494, July 2022.
 - [6] **Y. Xie**, T. Sun, L. Deng, and B. Chen, "Fast-robust Book Information Extraction System for Automated Intelligence Library," *AIID*, 2021.
 - [7] Z. Huang, **Y. Xie**, B. Ma, S. Shen, and J. Ma, "Distributed Potential Bayesian Game: Towards Fast and Scalable Multi-Hypothesis Game-theoretic Trajectory Planner," *Under Review*, 2025.
 - [8] J. Shi, **Y. Xie**, X. Chen, F. Wang, J. Chen, T. Rigen, T. Wo, and J. Ma, "HIMAT: Hierarchical Memory Fusion for Asymmetric Intention-Aware Multimodal Trajectory Generation," *Under Review*, 2025.
 - [9] H. Jiang, P. Qian, **Y. Xie**, X. Li, M. Liu, and J. Ma, "DuLoc: Life-Long Dual-Layer Localization in Changing and Dynamic Expansive Scenarios," *IROS (Robotics TOP)*, 2025.
 - [10] Y. Wang, Z. Peng, **Y. Xie**, Y. Li, H. Ghazzai, and J. Ma, "Learning the References of Online Model Predictive Control for Urban Self-Driving," *Under Review*, 2024.
 - [11] Y. Li, C. Zheng, K. Chen, **Y. Xie**, X. Tang, M. Y. Wang, and J. Ma, "Collision-Free Trajectory Optimization in Cluttered Environments Using Sums-of-Squares Programming," *RAL (Robotics TOP)*, 2024.
 - [12] X. Zhou, Z. Peng, **Y. Xie**, M. Liu, and J. Ma, "Game-Theoretic Driver Modeling and Decision-Making for Autonomous Driving with Temporal-Spatial Attention-Based Deep Q-Learning," *TIV*, 2024.